

Transit-Oriented Development (TOD) in the Age of AI, Climate Change and Aging Societies

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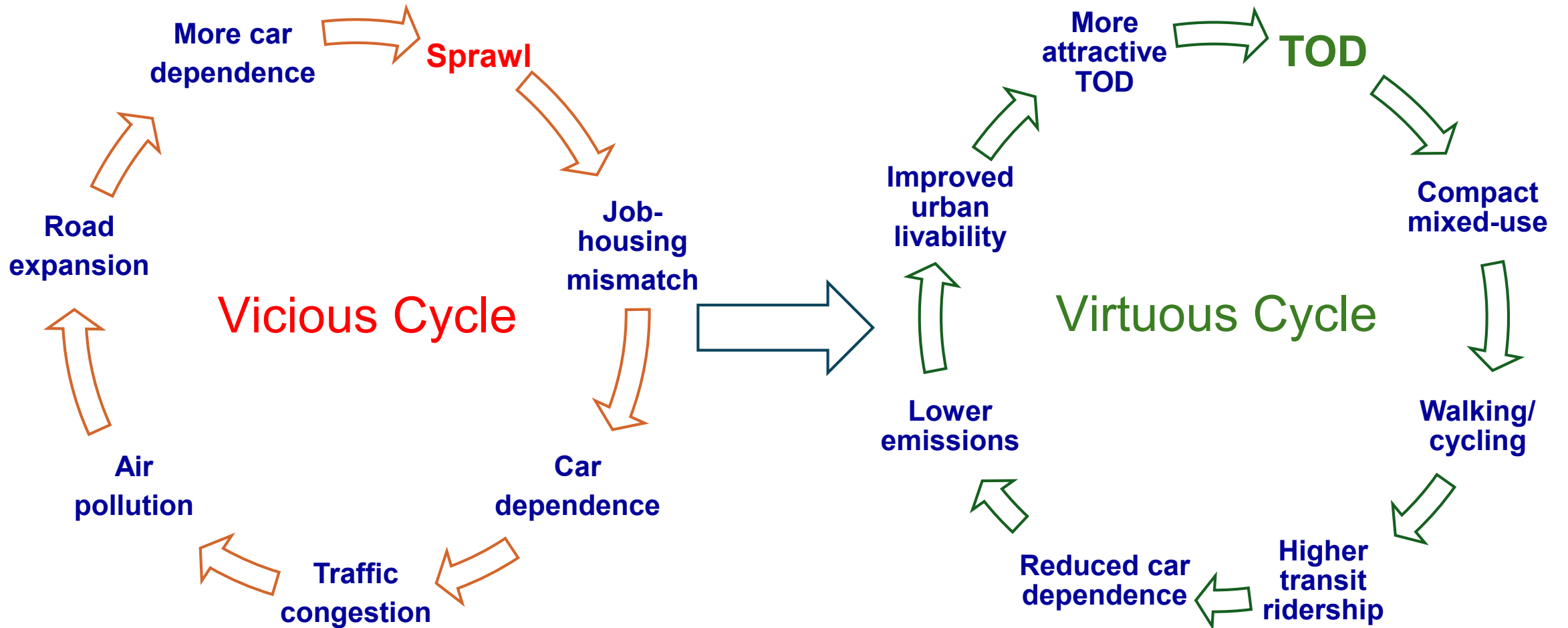


A blue pen with a silver tip is resting on a document that features a bar chart with blue bars of varying heights. The pen is positioned diagonally across the frame.

Outline

1. **Why does TOD matter today?**
2. **What is TOD?**
3. **Design Elements and Principles**
4. **Benefits and Barriers of TOD**
5. **Stakeholders in TOD**
6. **Evolution of TOD Thinking**
7. **Emerging Challenges and Future Directions**
8. **Concluding Remarks**

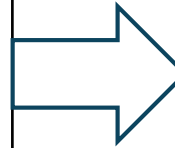
Why does TOD matter today?



Why does TOD matter today?

Yesterday's Challenges

- **Urban sprawl**
- **Traffic congestion**
- **Air pollution**
- **Energy consumption**
- **Parking demand**



Today's Challenges

- **Climate change**
- **Carbon neutrality**
- **Population aging**
- **Housing affordability**
- **Social equity**
- **AI / digital transformation**
- **Public health**
- **Urban resilience**

What is TOD?

- **There is no universally accepted definition of TOD.**
- **TOD is viewed quite differently from city to city.**
 - **A metropolitan city versus a middle-size city**
 - **North American cities versus Asian cities**
- **In many cities, TOD functions as a mixed-use activity center organized around a transit station.**

Relationship between TOD and the Car



- TODs are not necessarily **“anti car”**
- Instead, they make walking, cycling, and public transit more convenient and attractive alternatives to driving.

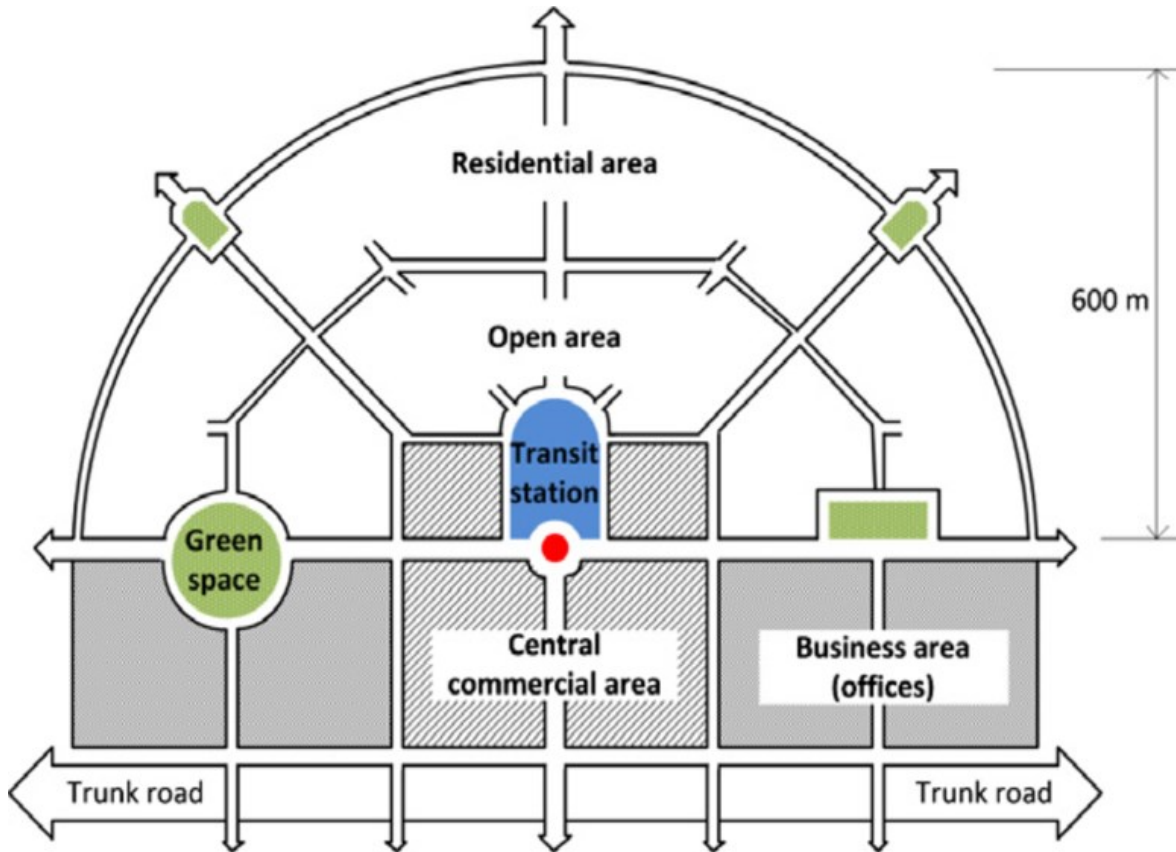
Contrast of TOD with AOD

**Automobile-
Oriented
Design (AOD)**



**Transit-
Oriented
Design (TOD)**

Design Elements



- **Transit Station**

- A walking radius of about 600-800 meters.

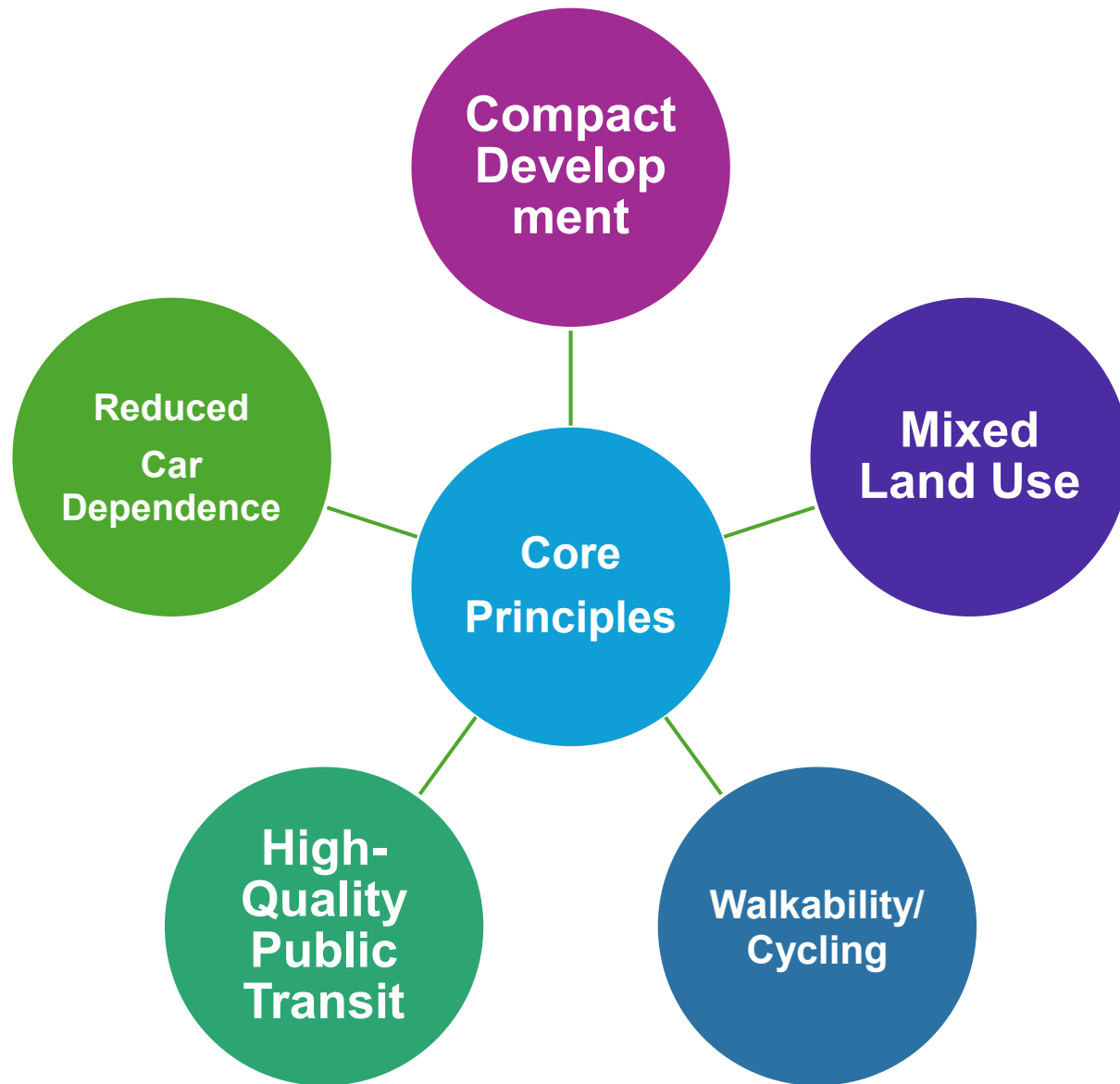
- **Mixed-use buildings**

- Housing, retail, offices.

- **Public services**

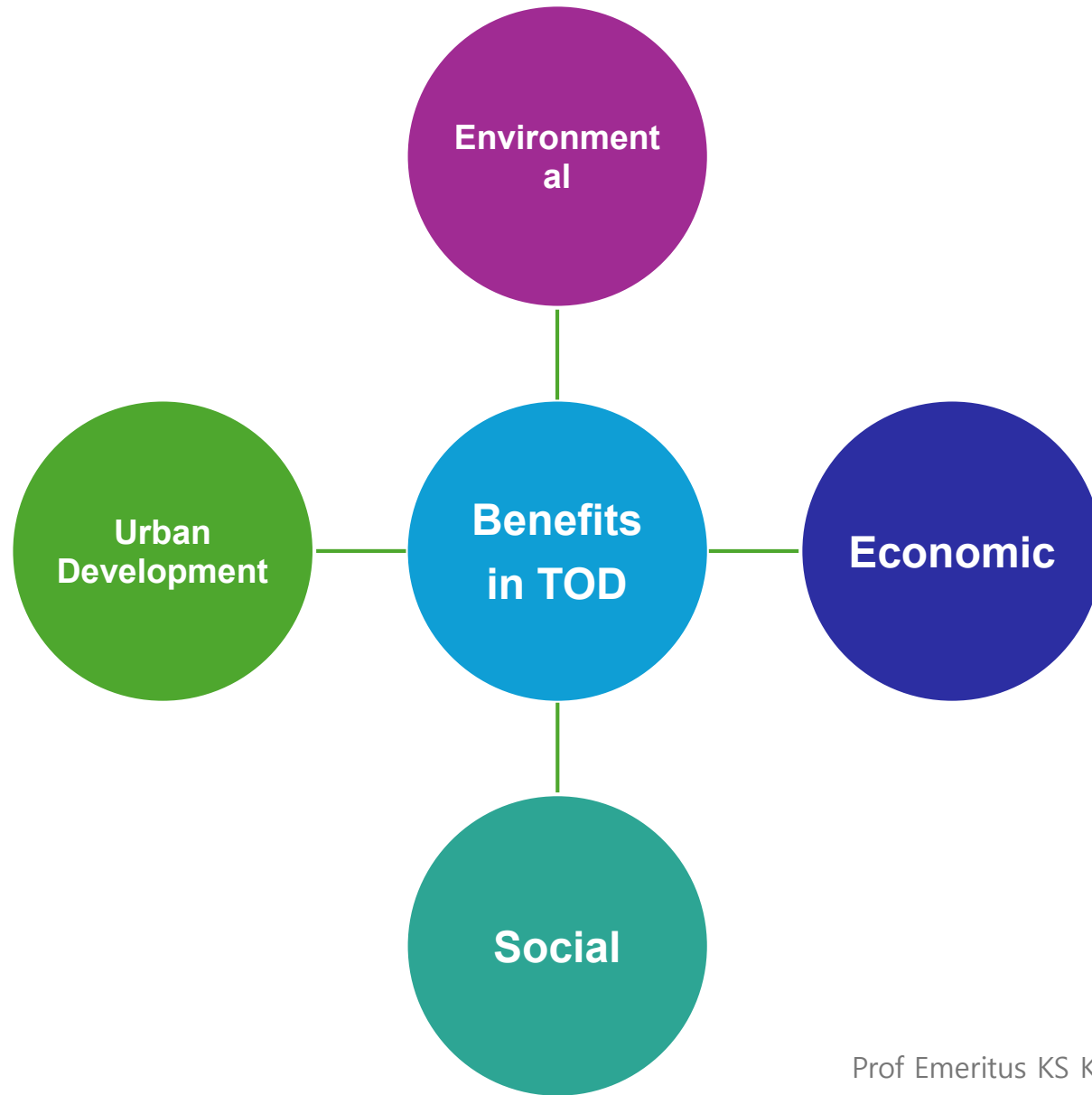
- Parks, schools, bike lanes, pedestrian streets.

Core Principles



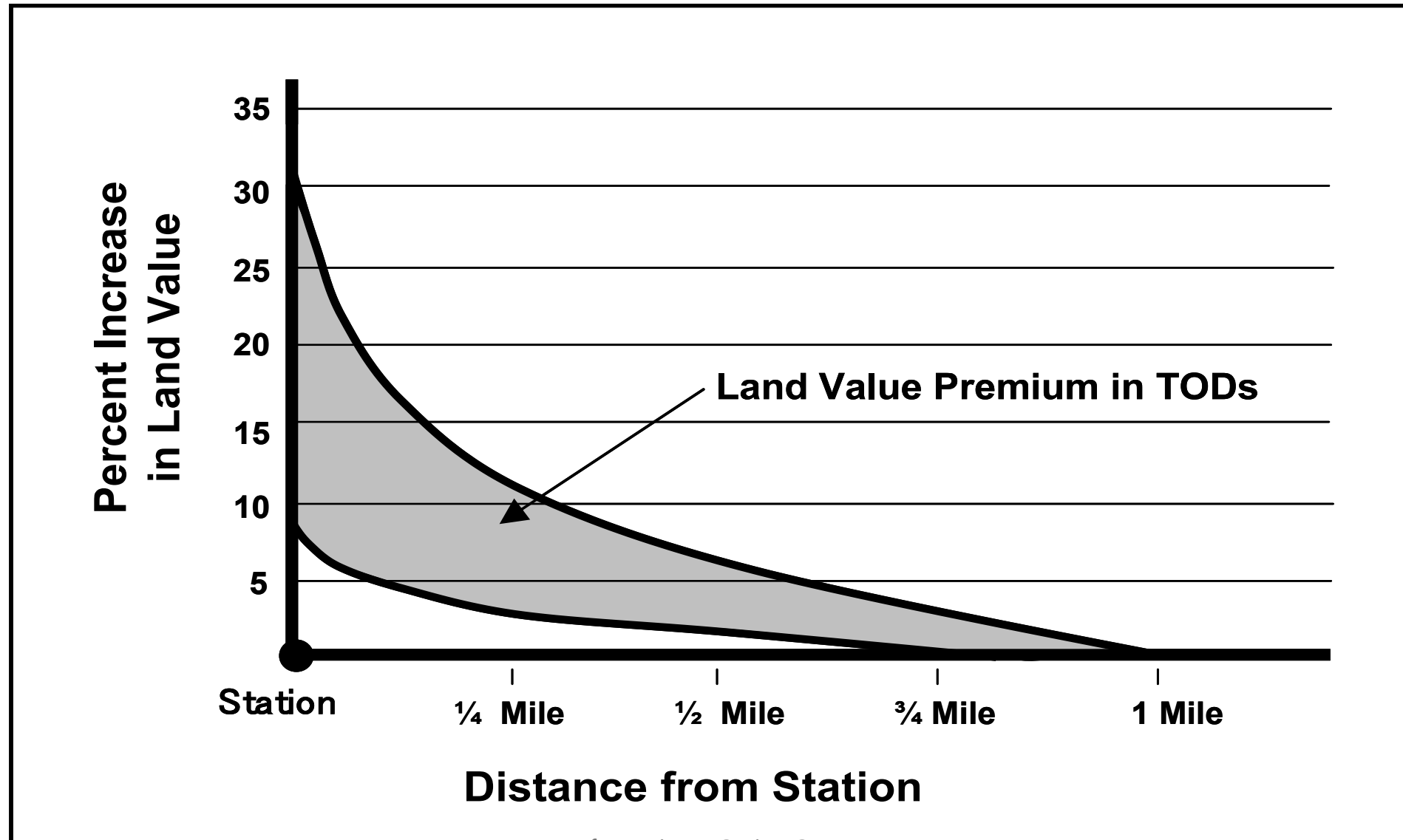
- **Moderate to high density, Efficient use of land**
- **Housing, Employment, Retail, Public services**
- **Safe pedestrian environment, Bicycle facilities, Universal accessibility**
- **Frequent, Reliable, Convenient, Multimodal integration**
- **Parking management, Traffic calming, Complete streets**

Key Benefits in TOD



- TOD contributes to more sustainable and climate-friendly cities.
- TOD enhance property values and revitalizes city centers.
- TOD improves quality of life.
- TOD increases urban livability.

Accessibility Benefits of TOD Land Value Premiums



Major Barriers to TOD



- **Institutions often change more slowly than cities.**
- **TOD requires long-term investment but often faces short-term financial constraints.**
- **Community acceptance is essential.**
- **NIMBY.**
- **TOD must be economically viable as well as socially desirable.**
- **Successful TOD requires capable institutions and skilled professionals.**

Key Stakeholders in TOD



Primary Interest

- Public good, Economic development
- Carbon reduction
- Economic vitality, Redevelopment
- Tax revenues
- Ridership, Service quality
- Financial sustainability
- Return on investment, Market demand
- Mixed-use development
- Affordability, Reliability
- Accessibility
- Neighborhood quality, Safety
- Affordable housing
- Digital twins, Smart ticketing
- Real-time information

Evolution of TOD Thinking

**Peter
Calthorpe**



**Robert
Cervero**



**Reid
Ewing**



**Ming
Zhang**



**The founder of the
TOD concept**

3Ds

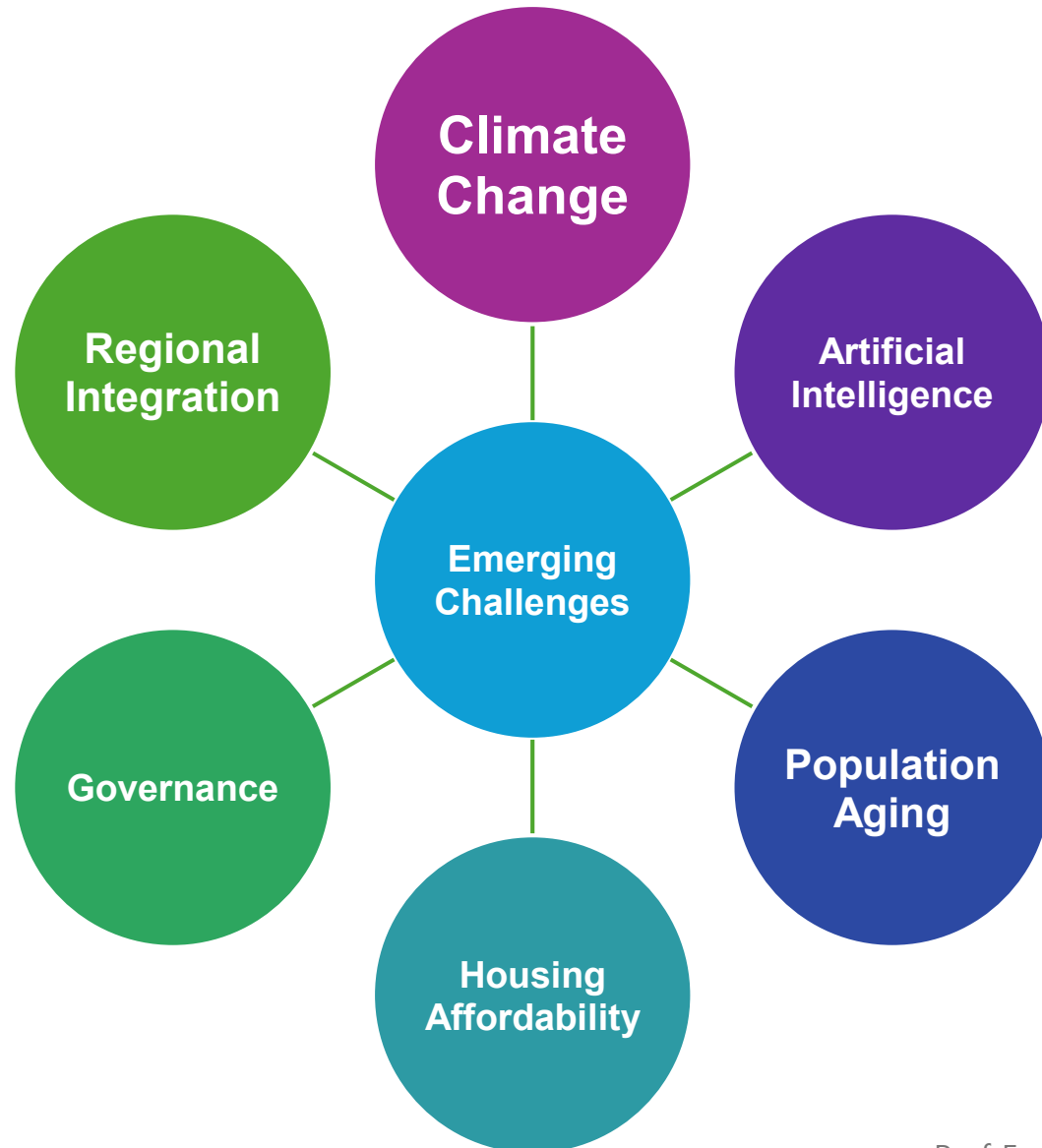
5Ds

**3Es + 2Es
Next-Gen TOD**

Scholars	Contribution		Year
Peter Calthorpe	- The pioneer of TOD - Design philosophy	- Mixed-use communities - Walkability	1993
Robert Cervero	- Empirical research 3Ds	- Travel demand - Transport-land use integration	1997
Reid Ewing Robert Cervero	- Expanded 5Ds - Travel behavior	- Quantitative indicators - Accessibility	2001
Ming Zhang	- Network perspective - Performance-based planning	3Es + 2Es - Adaptability - Participation	2025

Stage	Scholar	Core Question	Keywords
TOD 1.0	• Peter Calthorpe	• How should we design neighborhoods around transit?	• Design-oriented TOD
TOD 1.5	• Robert Cervero • Reid Ewing	• How does TOD influence travel behavior?	• Evidence-based TOD
TOD 2.0	• Ming Zhang	• How should TOD respond to contemporary urban challenges?	• Performance-based, network-oriented TOD
TOD 3.0?	• Future	• How can TOD create intelligent, resilient, and age-friendly cities?	• Future vision

Emerging Challenges



Keywords

- **Net-zero development, Green infrastructure**
- **AI-supported scenario analysis, Digital twins, Real-time mobility**
- **Universal Design, Barrier-free stations, Healthy walking environments, Intergenerational communities**
- **Affordable housing, Social equity, Inclusive TOD**
- **PPP, Citizen participation, Cross-sector collaboration, Data sharing**
- **TOD → TOC → TOR**

Strategic Directions for the Future of TOD

Directions	Keywords
1. Think beyond individual stations	• From TOD to Transit-Oriented Corridors and ultimately Transit-Oriented Regions.
2. Put people before infrastructure	• Creating walkable, inclusive, healthy and age-friendly communities.
3. Make TOD more inclusive	• Serving people of all ages, incomes, and abilities.
4. Harness AI and digital technologies	• Improving planning, mobility, and transit operations.
5. Strengthen governance and financing	• Collaboration, innovation, and long-term investment.
6. Build climate-resilient regions	• Low-carbon, energy-efficient, and prepared for future challenges.

Concluding Remarks

- Transit-Oriented Development is not a panacea.
- It cannot solve every urban problem.
- However, when implemented at scale and integrated into regional planning, TOD can become one of the most powerful strategies for creating sustainable, inclusive, and resilient cities.
- The future of TOD is not just about connecting people to transit.
- It is about connecting people to opportunities, sustainability, and a better quality of life.
- In the age of AI, climate change, and population aging, the next generation of TOD should become the foundation for intelligent, resilient, inclusive, and healthy cities.





**Thank
You**